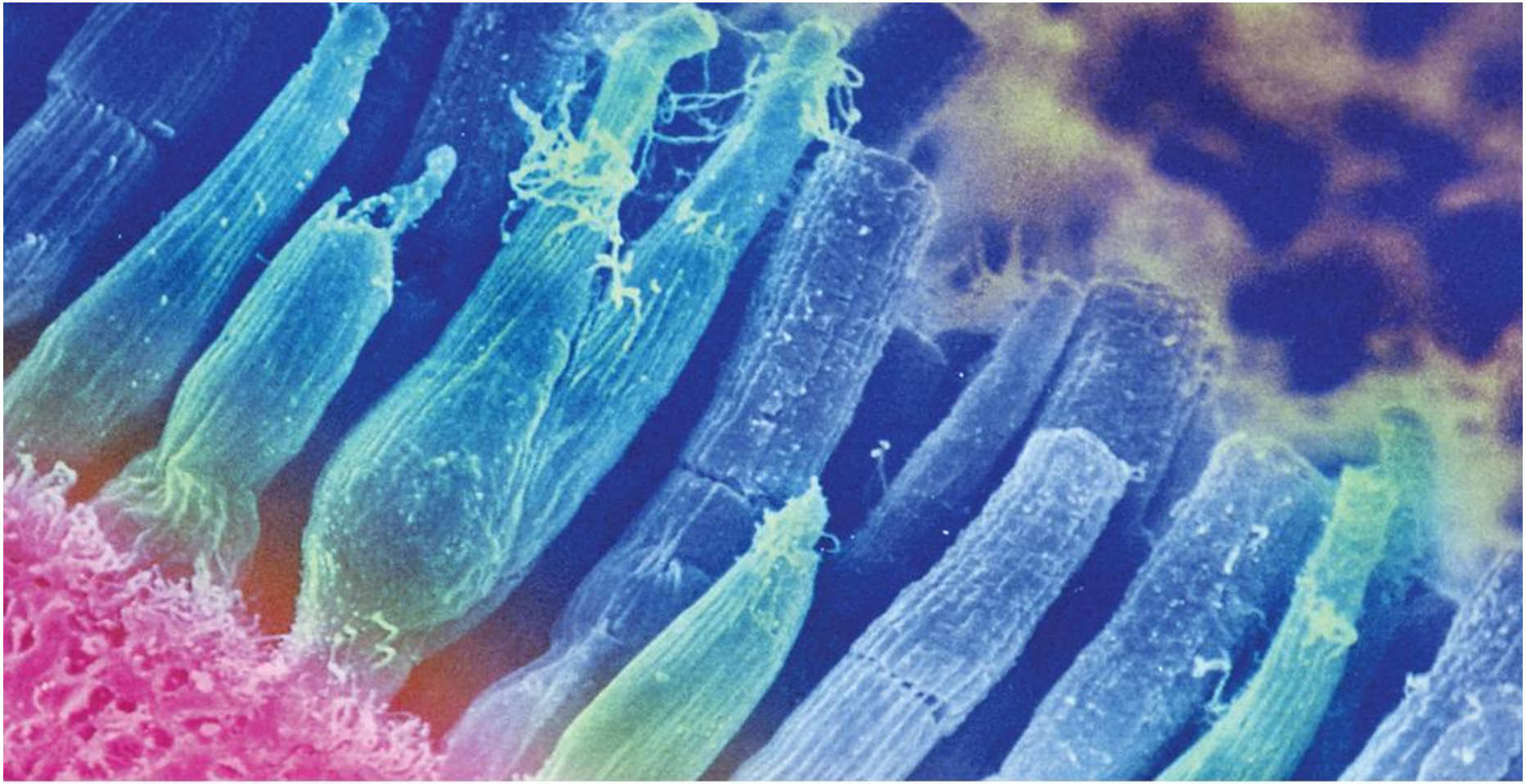


Chapter 6. The Peripheral Nervous System: Afferent Division and Special Senses

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A scanning electron microscope of rods and cons



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Peripheral nervous system in terms of homeostasis

- The nervous system consists of CNS and PNS

- PNS is composed of the afferent and efferent fibers that relay signals between the CNS and the periphery (other parts of the body)

- The afferent division transmits signals to the CNS, informing about both the internal and external environment
- Afferent input makes the CNS know what is going on and is also used to plan for voluntary movements unrelated to homeostasis

- Then, the CNS makes the adjustments to maintain homeostasis by controlling the activities of effector organs (e.g. muscles and glands), by transmitting signals from the CNS through the effector division

6.1 Receptor Physiology

Receptor physiology

- PNS = nerve fibers that carry information between the CNS and periphery (other parts of the body)
- Stimulus = change detectable by the body
  - Heat, light, sound, pressure, chemical changes

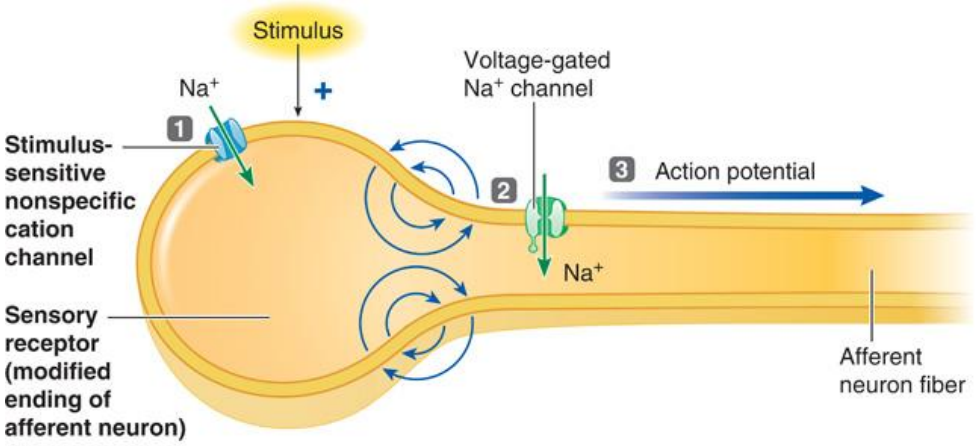
- Afferent neurons have sensory receptors (or receptors) at their peripheral endings
- Stimuli bring graded potential known as receptor potentials, in turn triggering action potential in the afferent fiber
- Sensory transduction = conversion of stimulus energy into a receptor potential

Receptors have differential sensitivities to stimuli

- Types of receptors according to their adequate stimulus
  - Photoreceptors (visible light)
  - Thermoreceptors (heat and cold)
  - Osmoreceptors (solute concentration in the ECF and changes in osmotic activity)
  - Mechanoreceptors (mechanical energy)
  - Chemoreceptors (specific chemicals)
  - Nociceptors or pain receptors (pain, tissue damage)

- A stimulus alters the receptor's permeability, leading to a graded potential
  - Graded potential revisited: triggering event opens ion channel, usually permitting the entry of Na^+
  - Receptor potential = local depolarization-mediated graded potential (except for photoreceptor, hyperpolarization)
  - The stronger the stimulus, the greater the receptor potential is

Conversion of receptor potential into action potential

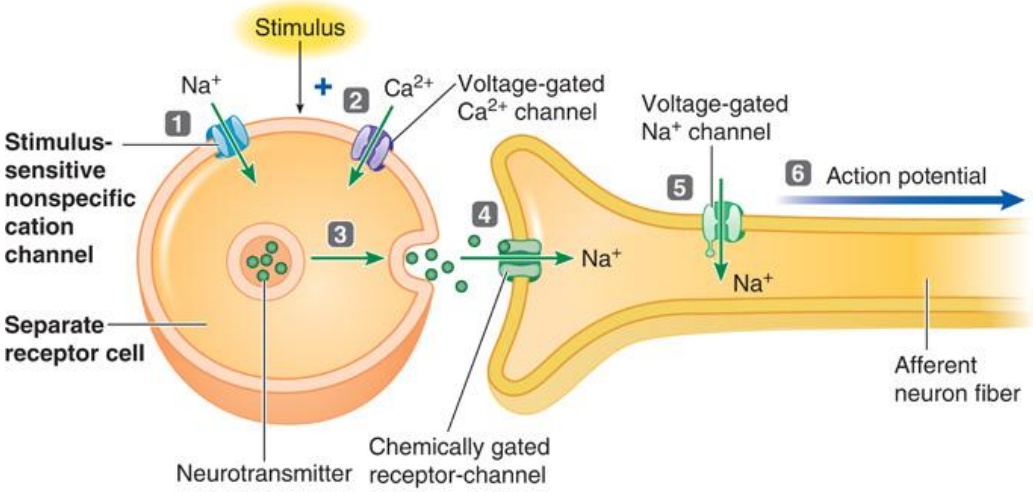


1 In sensory receptors that are specialized afferent neuron endings, stimulus opens stimulus-sensitive channels, permitting net Na^+ entry that produces receptor potential.

2 Local current flow between depolarized receptor ending and adjacent region opens voltage-gated Na^+ channels.

3 Na^+ entry initiates action potential in afferent fiber that self-propagates to CNS.

(a) Receptor potential in specialized afferent ending: receptors that directly generate action potentials



1 In sensory receptors that are separate cells, stimulus opens stimulus-sensitive channels, permitting net Na^+ entry that produces receptor potential.

2 This local depolarization opens voltage-gated Ca^{2+} channels.

3 Ca^{2+} entry triggers exocytosis of neurotransmitter.

4 Neurotransmitter binding opens chemically gated receptor-channels at afferent ending, permitting net Na^+ entry.

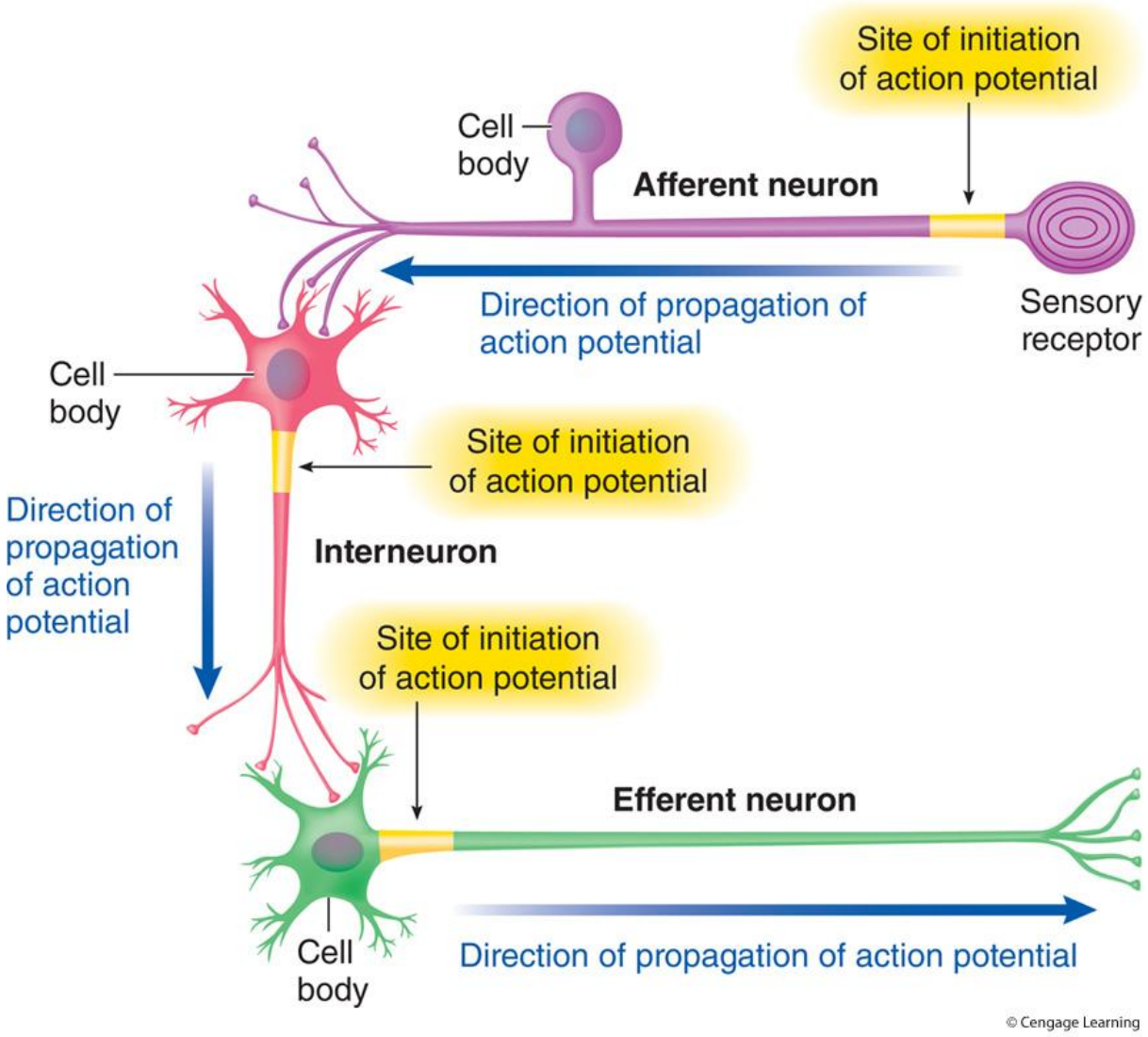
5 Resultant depolarization opens voltage-gated Na^+ channels in adjacent region.

6 Na^+ entry initiates action potential in afferent fiber that self-propagates to CNS.

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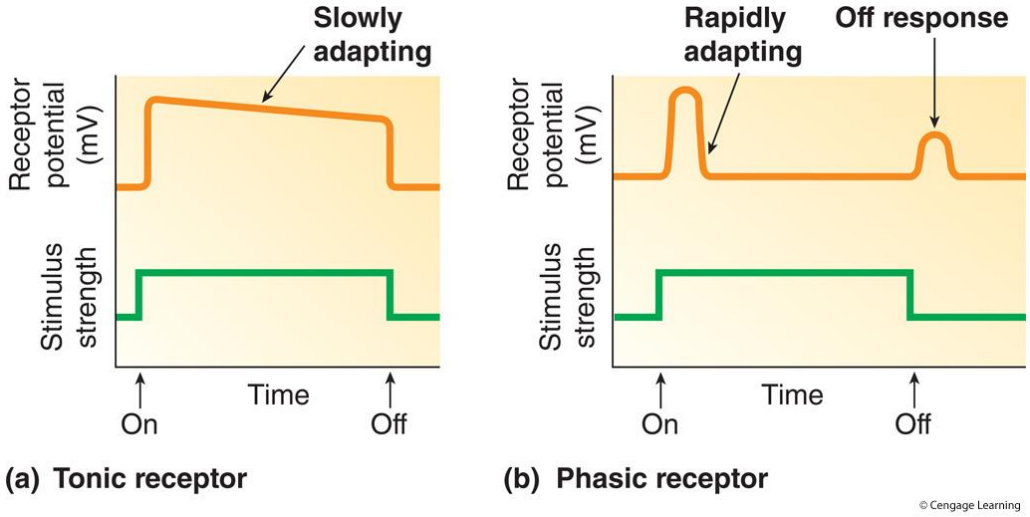
(b) Receptor potential in separate receptor cell: receptors that release neurotransmitters to activate afferent neurons

Action potential initiation site in three neuron types



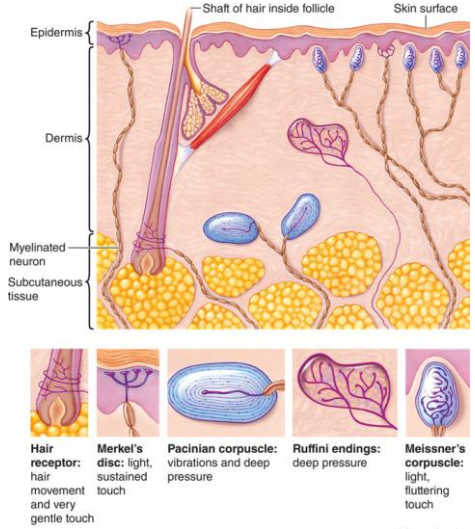
Receptors adapt slow or fast to sustained stimulation

- Depending on their speed of adaptation
  - Tonic receptors
    - Do not adapt or adapt slowly
    - Useful to maintain information about stimulus
    - Muscle stretch receptor, joint proprioceptor
    - Keep generating action potentials
    - Good for monitoring constant conditions like posture or pain
  - Phasic receptors
    - Rapidly adapting, by no longer responding to a maintained stimulus
    - Useful to signal a change in stimulus intensity
    - Pacinian corpuscles, tactile (touch) receptors
    - When stimulus is removed, short off response
    - Good for signaling a change in stimulus intensity



Tactile receptors

- Mechanoreceptors in the skin
- Mechanical forces open non-specific cation channels, leading to Na^+ entry and receptor potential increase
  - Hair receptor: rapidly adapting, senses hair movement and gentle touch
  - Merkel's disc: slowly adapting, detects light and sustained touch
  - Pacinian corpuscle: rapidly adapting, senses vibration and deep pressure
  - Ruffini endings: slowly adapting, senses deep, sustained pressure, skin stretching
  - Meissner's corpuscle: rapidly adapting, sensitive to light and tickling



Somatosensory pathways

- Visceral afferent (내장구심성)
  - Incoming pathway for information derived from the organs inside the body cavity
  - Carries subconscious (잠재의식, 자기도 모르는) input from internal organ
- Sensory afferent (감각구심성)
  - Incoming pathway for sensory information, inputs from receptors located at the skin or in the muscles, tendon, joints, and special sense organs
  - Carries conscious (의식하는) sensory information from somatic and special sensory organs
    - Somatic (body) sense: body surface like skin and proprioception (muscle, joint)_touch, pressure, vibration, temperature, pain, proprioception
    - Special senses: vision (retina), hearing (organ of Corti), smell (olfactory epithelium), taste (taste buds), equilibrium/balance (vestibular apparatus of inner ear)
  - Also provide input for subconscious processing, including proprioceptive control of posture and movement

6.2 Pain

Pain

- Primarily a protective mechanism triggered on stimulation of danger-sensing nociceptors
- Brings to conscious awareness tissue damage that is occurring or about to occur
- Nociceptors do not adapt to sustained or repetitive stimulation
- Storage of painful experiences in memory helps us avoid potentially harmful events in the future

Nociceptors

- Stimulation of nociceptors enables the perception of pain
- Pain sensation is accompanied by motivational and emotional responses
- Categories of pain receptors
  - Mechanical nociceptors: mechanical damage (cutting, crushing, pinching)
  - Thermal nociceptors: temperature extremes
  - Polymodal nociceptors: all kinds of damaging stimuli (irritating chemicals from tissue injury)
- Afferent pain fibers
  - Fast pain fiber (delta fiber) and slow pain fiber (C fiber)

Nociceptors

- Stimulation of nociceptors enables the perception of pain
- Pain sensation is accompanied by motivational and emotional responses
- Categories of pain receptors
  - Mechanical nociceptors: mechanical damage (cutting, crushing, pinching)
  - Thermal nociceptors: temperature extremes
  - Polymodal nociceptors: all kinds of damaging stimuli (irritating chemicals from tissue injury)
- Afferent pain fibers
  - Fast pain fiber (delta fiber): stimulation of mechanical and thermal nociceptors, myelinated -> sharp and fricking pain
  - Slow pain fiber (C fiber): stimulation of polymodal nociceptors, unmyelinated, -> dull, aching, and burning pain
- NTs produced by afferent pain fibers: substance P and glutamate

6.3 Eye: Vision

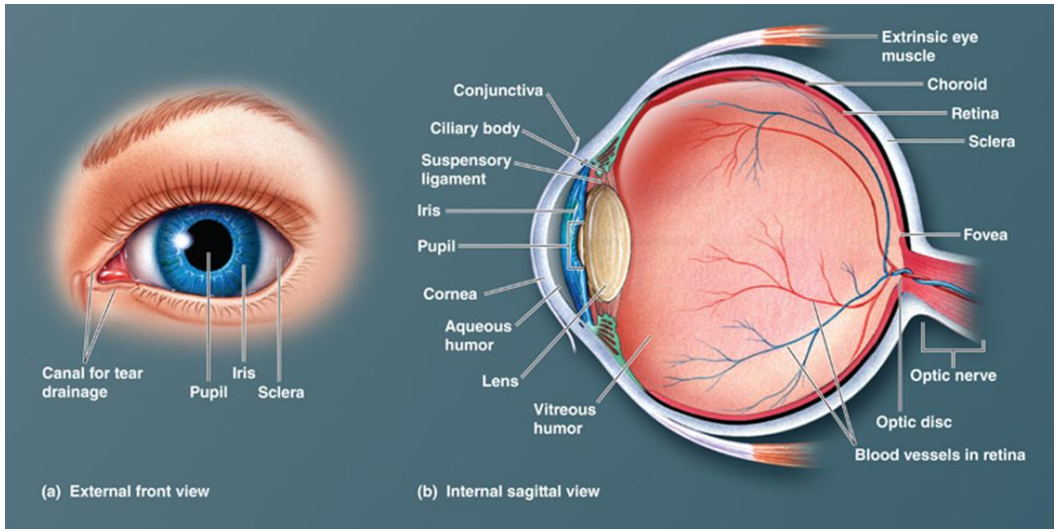
Eyes

- Eyes capture patterns of light from the environment
- These patterns form an image on the retina, which contains light-sensitive cells
- Coded image on the retina is transmitted through visual processing steps until it is finally consciously perceived

- Eyes have a protective mechanism and layers
  - Eyeball is sheltered by bony socket
  - Eyelids: act like shutters to protect exposed part
  - Tears produced by lacrimal gland: lubricating, cleansing, bacteria-killing
  - Eyelashes: trap dust

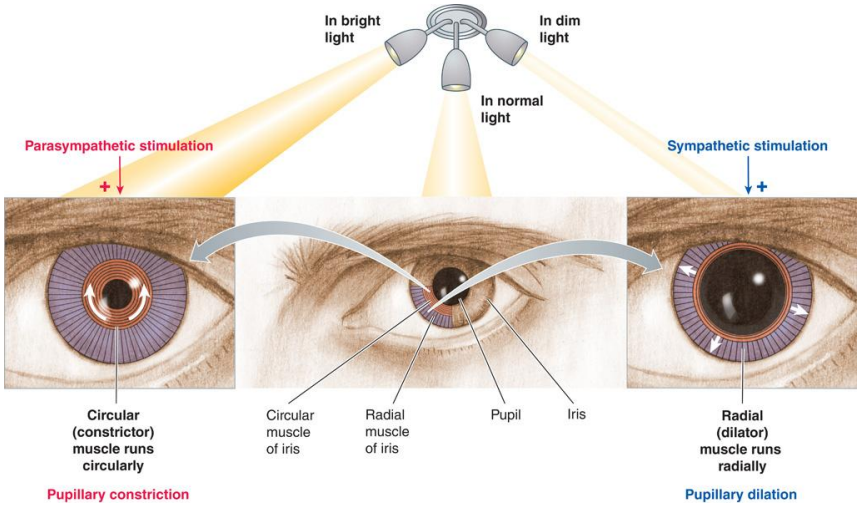
The eye is a fluid-filled sphere enclosed by three layers

- Outer layer
  - Sclera (visible white part) and cornea (transparent outer layer)
- Middle layer
  - Choroid (pigmented): contains blood vessels nourishing retina
  - Ciliary body: produces aqueous humor and controls lens shape for focusing
  - Iris: the colored part regulating the amount of light entering through pupil
- Innermost layer
  - Retina: outer pigmented layer + inner nervous-tissue layer (rods and cons)
- Two fluid-filled cavities
  - Vitreous humor
  - Aqueous humor
  - Separated by lens



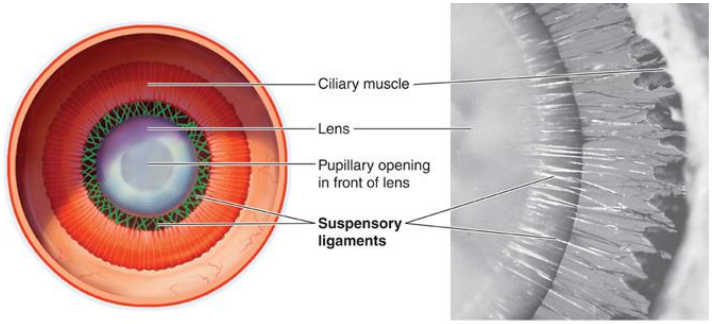
Eyes process light

- The amount of light entering the eye is controlled by the iris
- Iris: thin, pigmented, smooth muscle
- The pupil is the opening at the center of the iris through which light passes
- Iris forms a visible ringlike structure within the aqueous humor
  - Iris muscles are controlled by the autonomic nervous system
  - Muscle fibers shorten when they contract
  - Under bright light, parasympathetic nervous system triggers circular muscles contraction
  - Under dim light, sympathetic nervous system triggers radial muscles contraction

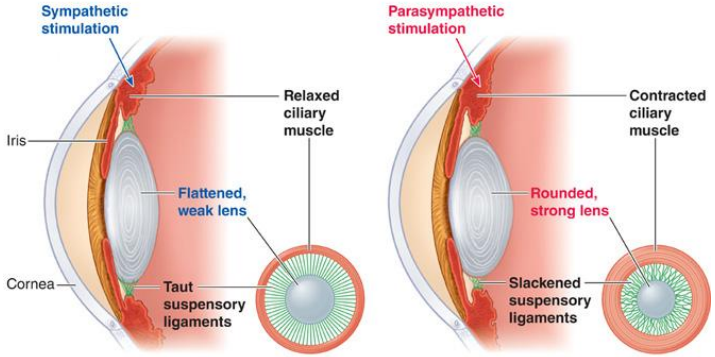


Eyes increase the strength of the lens for near vision

- Accommodation: the ability to adjust the strength of lens
  - Increases the strength/thickness of the lens for near vision
  - The strength of the lens depends on its shape, regulated by ciliary muscle
  - Suspensory ligament: extends from ciliary muscle to the outer edge of lens
  - When ciliary muscle are relaxed, suspensory ligaments are tight, more tension on the lens, flattening it
  - When ciliary muscle are contracted, suspensory ligaments are loose, less tension on the lens, rounding it
  - Round lens -> near vision



(a) Anterior view of suspensory ligaments extending from ciliary muscles to lens



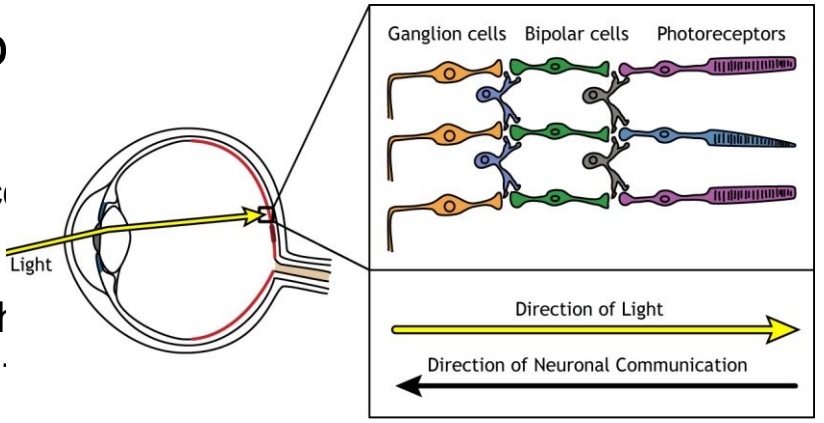
(b) Sagittal and anterior view when ciliary muscle is relaxed

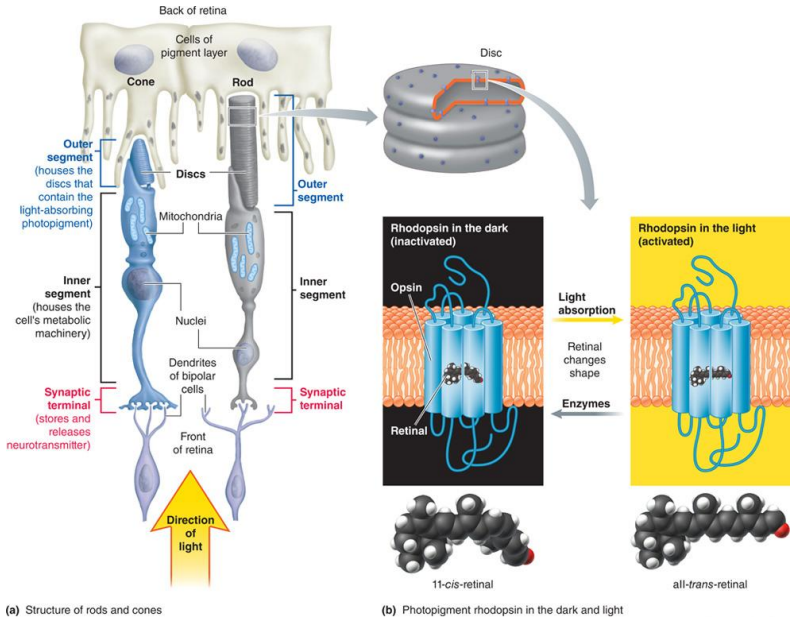
(c) Sagittal and anterior view during accommodation, when ciliary muscle is contracted

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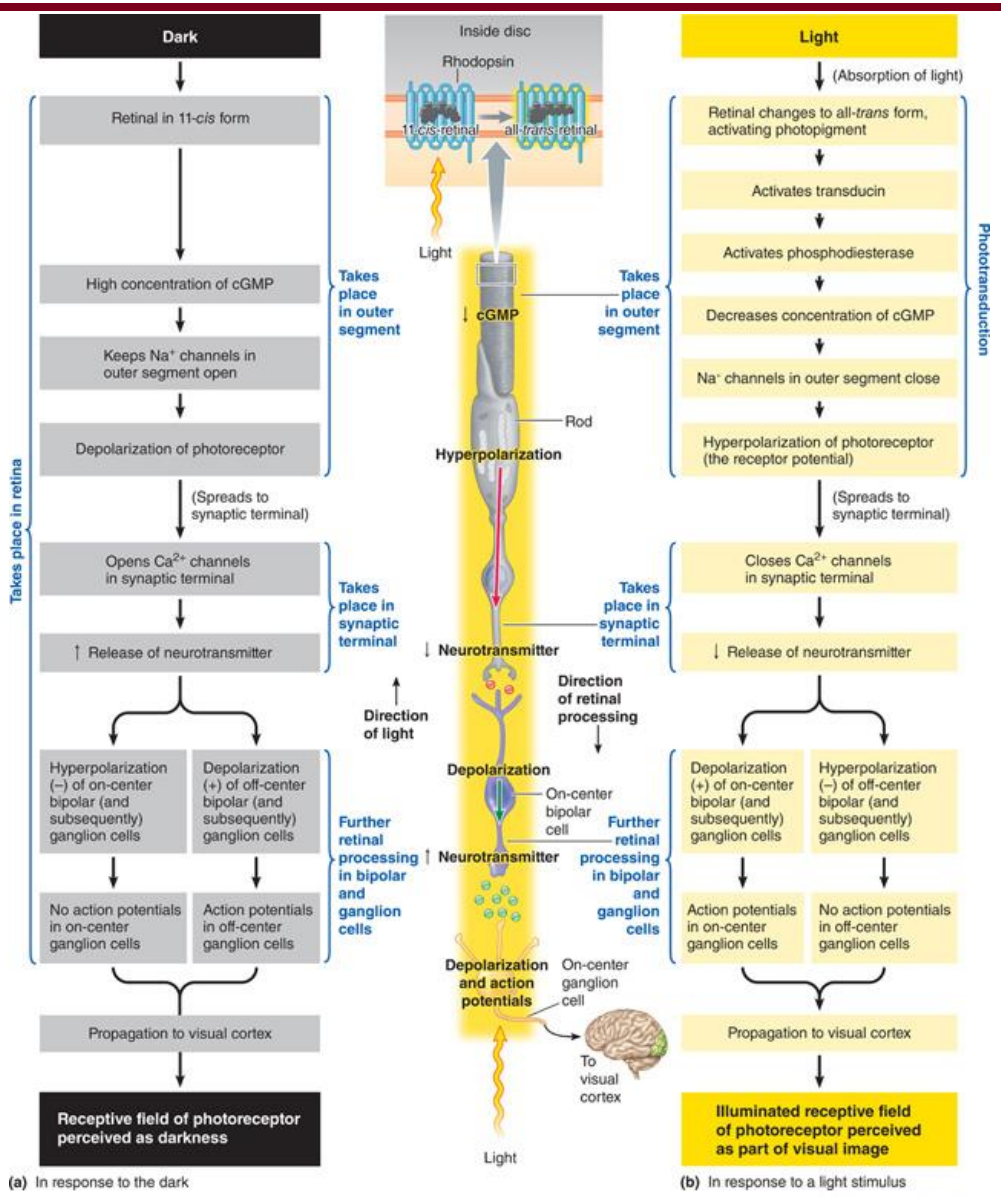
Phototransduction by retinal cells

- Phototransduction by retinal cells co
  neural signals
  - Photoreceptors (=rods and cone cells) c
    - Outer segment: contains light-absorbing
    - Inner segment: metabolic machinery of t
    - Synaptic terminal: stores and releases N
  - Photopigments
    - light-sensitive substances
    - found in retinal rods and cones
    - absorb light energy and convert it into electrical signals, initiating vision
    - retinal undergoes structural change
    - hyperpolarization of photoreceptors
    - action potential in retinal ganglion cells
    - visual perception in the brain
    - decreased release of glutamate





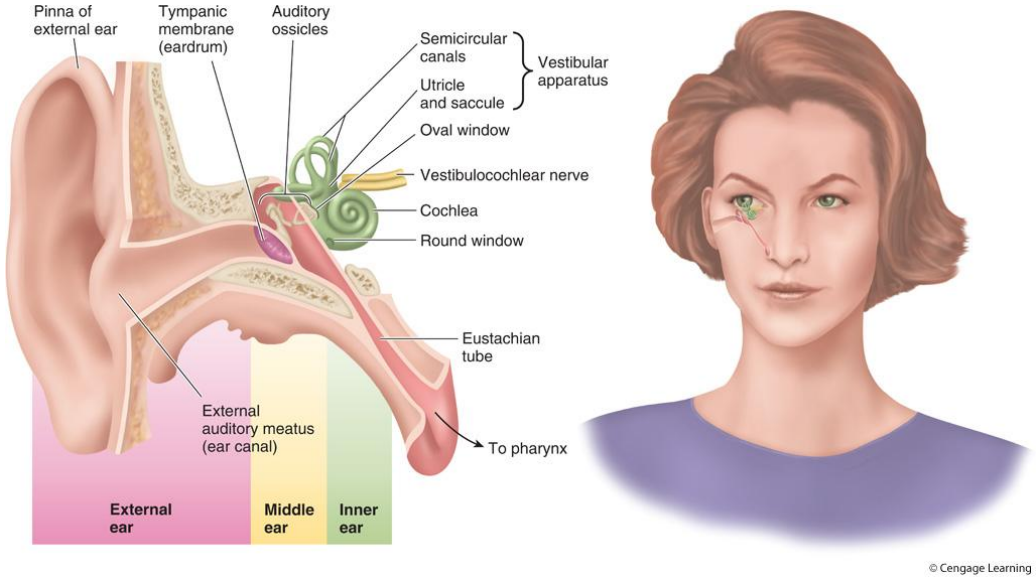
Phototransduction by retinal cells



6.4 Ear: Hearing and Equilibrium

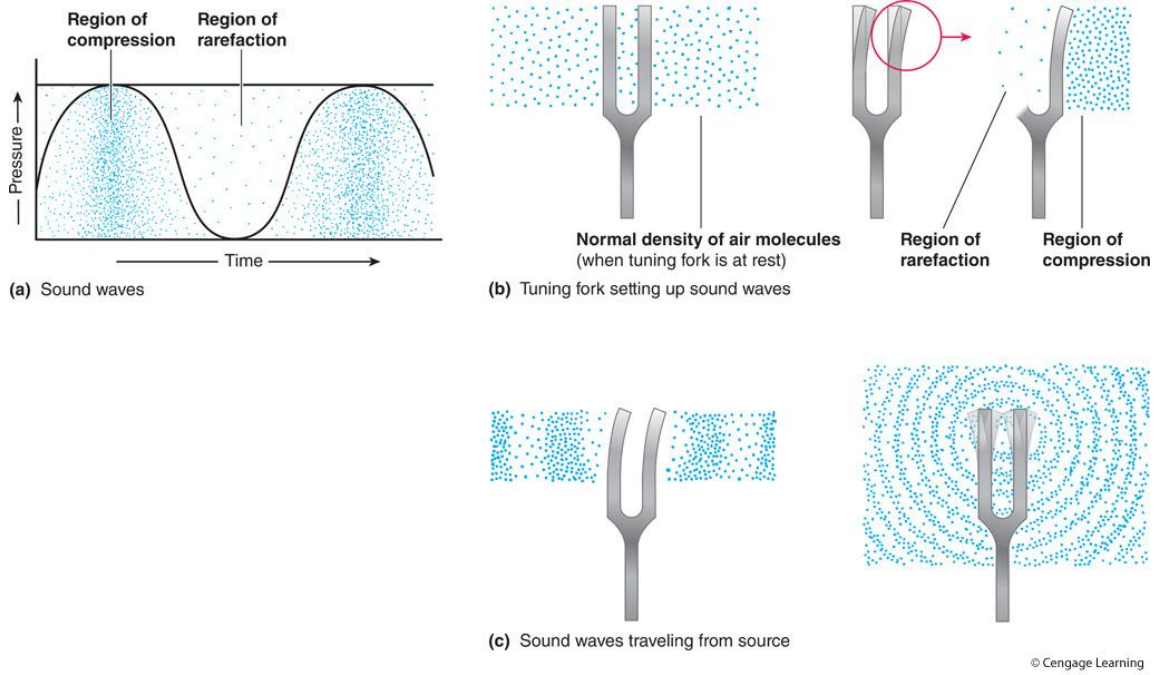
Overview of the ear

- Ear consists of three parts
  - External ear, the middle ear, and the inner ear (cochlea, vestibular apparatus)
- External and middle ear amplify sound energy and transmit airborne sound waves to the fluid-filled inner ear



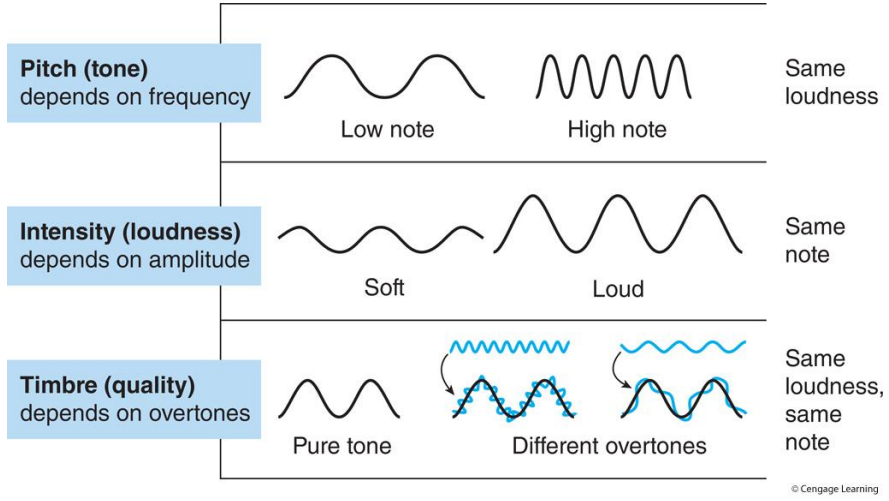
Sound waves

- Hearing: neural perception of sound energy
- Sound waves
  - Traveling vibrations of air
  - Made of areas where air is pushed together (compression) and areas where air is spread apart (rarefaction)



Sound waves

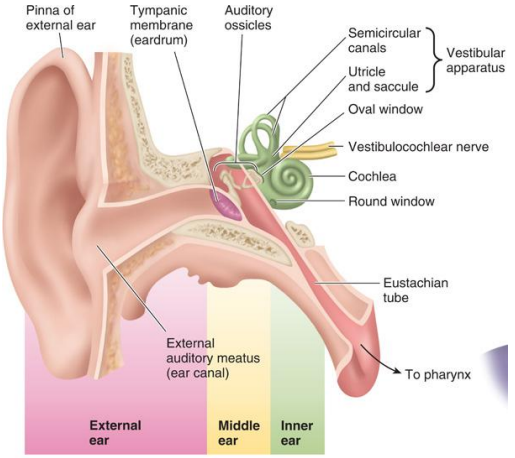
- Sound waves can also travel through media other than air, like water
  - More energy is required since fluid has higher resistance to change (inertia)
- Sound characteristics
  - Pitch (tone): *frequency* of vibrations
  - Intensity (loudness): *amplitude* of sound waves
  - Timbre (quality): overtone



External ear

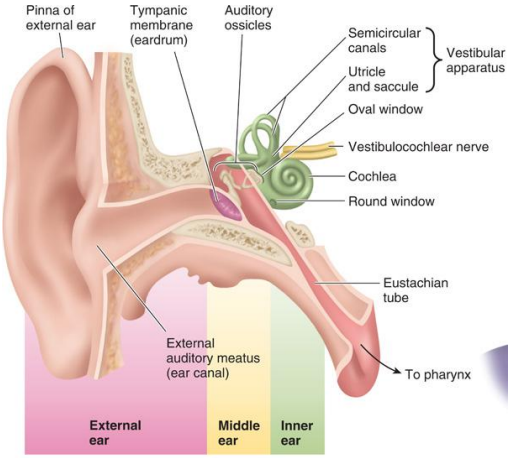
- The external ear plays a role in sound localization
  - Pinna (ear)
  - External auditory meatus (ear canal)
  - Tympanic membrane (eardrum)

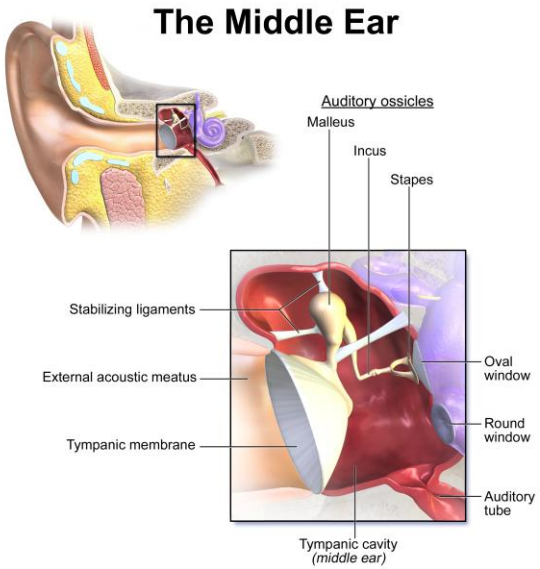
- Tympanic membrane vibrates in response to the sound waves, converting sound energy into mechanical vibrations



Middle ear

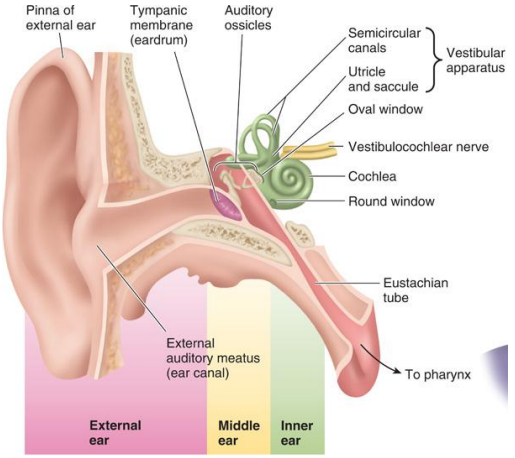
- The bones in the middle ear convert vibrations into fluid movement in the inner ear
- Movable chain of three bones (auditory ossicles)
  - Malleus
  - Incus
  - stapes

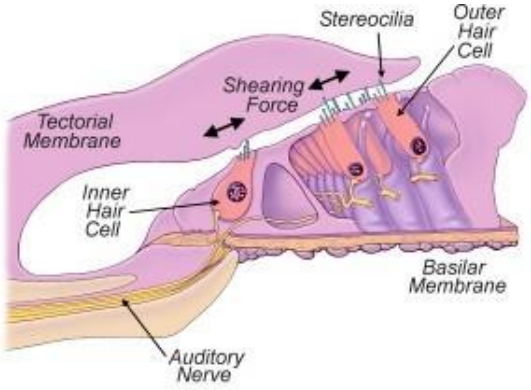




Inner ear

- Cochlea
  - Hearing portion of the inner ear
  - *Fluid*-filled cavity converting vibrations into electrical signals
  - Pea-sized, snail-shaped
  - Organ of Corti containing hair cells
- Organ of Corti
  - Sense organ for hearing
  - Hair cells here transduce *fluid* movement into electrical signals
    - Inner hair cells: convert fluid vibration into the electrical impulses
    - Outer hair cells: do not signal the brain, but act as micromechanical amplifier giving sensitivity to sound intensity and discrimination

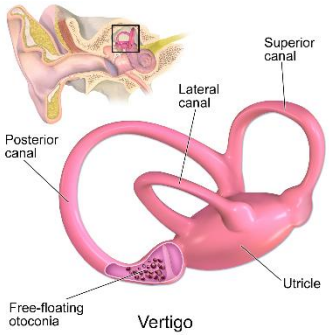


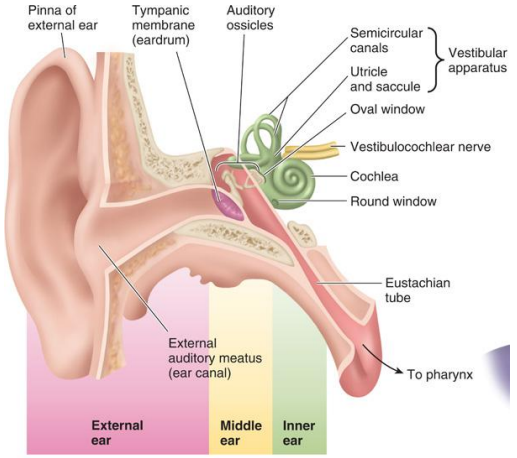


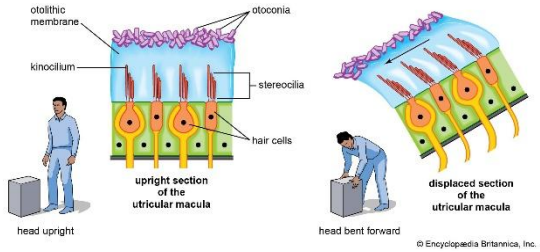
The Organ of Corti

Inner ear

- Vestibular apparatus
  - Provides information essential for the sense of equilibrium (평형감각) and coordinating head movements with eye (vestibulo-ocular reflex, 전정안구반사) and with posture (vestibulospinal reflex, 전정척수반사)
  - Semicircular canals
    - three interconnected, fluid-filled tubes
    - lateral, superior, posterior
    - detect rotational head movement and rotation acceleration
  - Otolith organs
    - Utricle (타원낭) and saccule (구형낭)
    - Detect linear acceleration, gravitational forces, and head orientation, by sensing otoliths (calcium carbonate crystals)







6.5 Chemical Senses: Taste and Smell

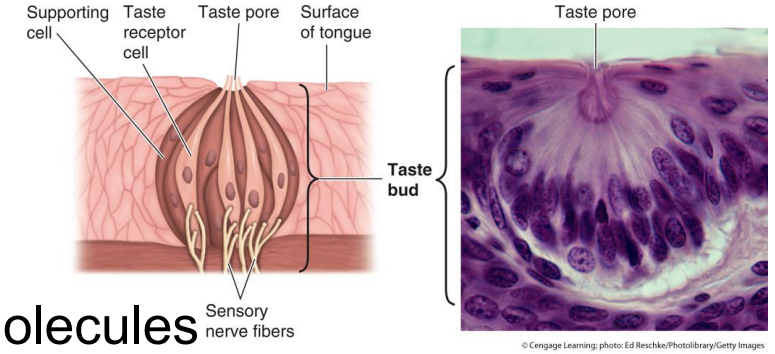
Chemical senses: taste and smell

- Eyes: photoreceptor
- Ears: mechanoreceptor
- Taste and smell receptor: chemoreceptor
  - Generate neural signals upon binding with particular chemical
  - Their sensation induces flow of digestive juices and affect appetite
  - Quality checkpoint for substances before ingestion

- Gustation: taste
- Olfaction: smell

Chemical senses: taste

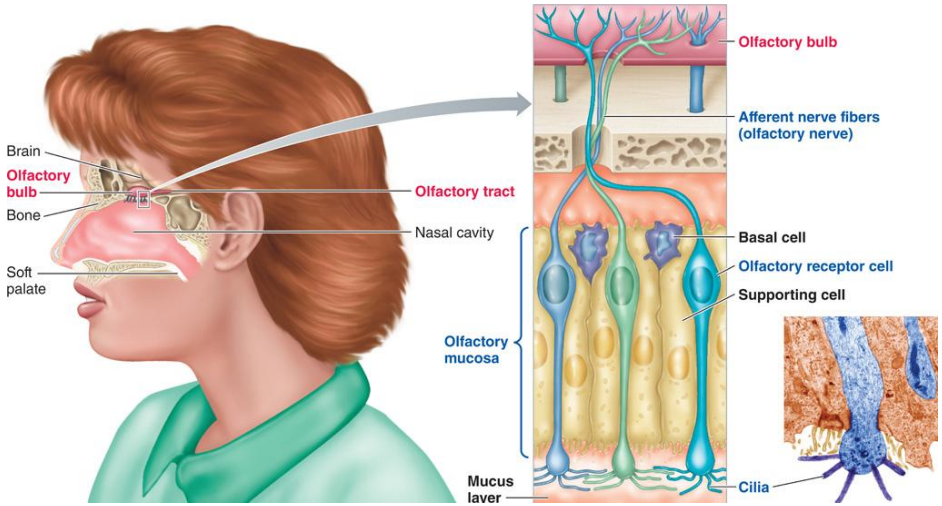
- Taste receptor cells are located primarily within tongue taste buds
- A taste bud consists of about 50 long, spindle-shaped *taste receptor cells* and *supporting cells* in an arrangement
- Taste pore: opening at the top
  - Fluids in the mouth come into here
  - Contact with surface of receptor cells
  - Plasma membrane of taste receptor cells contain receptors for chemical molecules

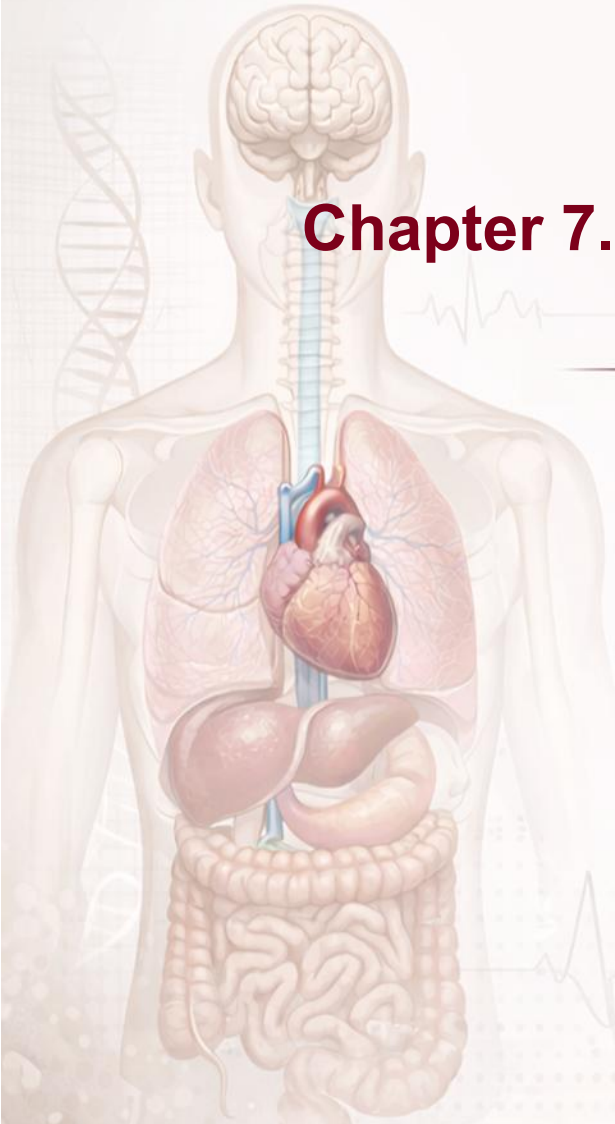


  -> depolarizes receptor potential -> opens V-gated Ca^{2+} channels -> NT release -> generates action potential in sensory nerve fibers
- Taste discrimination is coded by patterns of activity in various taste bud receptors
  - Salty, sour, sweet, bitter, and umami tastes
  - Brain stem -> thalamus -> primary gustation cortex

Chemical senses: smell

- The olfactory receptors in the nose are specialized endings of renewable afferent neurons
- Three types of cells
  - Olfactory receptor cells
    - Detects odor
    - Afferent neuron whose receptor portion lies in the olfactory mucosa and afferent axon portion traverses into the brain via olfactory nerve
  - Supporting cells: secrete mucus, coating the nasal passage
  - Basal cells: precursor for new olfactory receptor cells, replaced every 2 mo.





Chapter 7. The Peripheral Nervous System: Efferent Division

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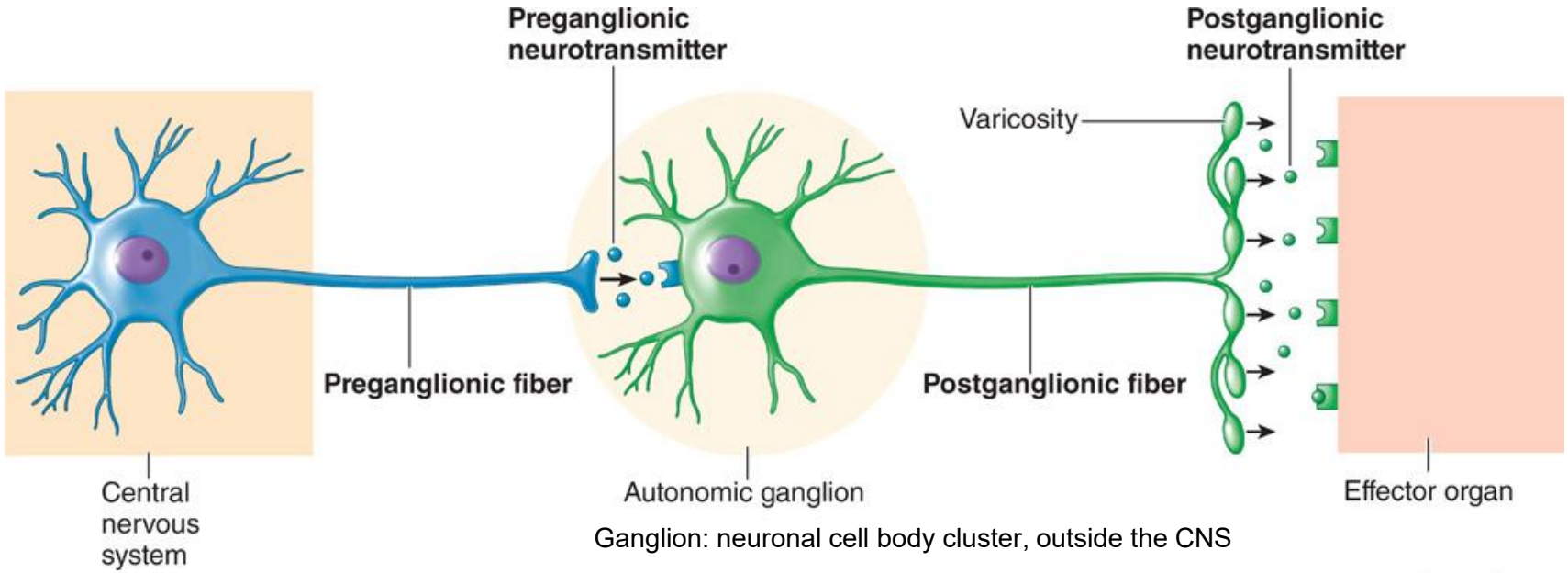
7.1 Autonomic Nervous System

Autonomic nervous system

- Efferent division of the PNS
  - It is a communication link where the CNS controls muscles and glands, the effectors that carry out intended effects or actions
  - CNS regulates the effectors by initiating APs in the cell bodies of efferent neurons whose axon terminates in effectors

- Autonomic nervous system
  - Involuntary control
  - Cardiac muscle, smooth muscle, most exocrine glands, some endocrine glands, and adipose tissue are innervated by ANS
- Somatic nervous system
  - Voluntary control
  - Skeletal muscle is innervated by PNS

Pathways and division of the ANS



Ganglion: neuronal cell body cluster, outside the CNS

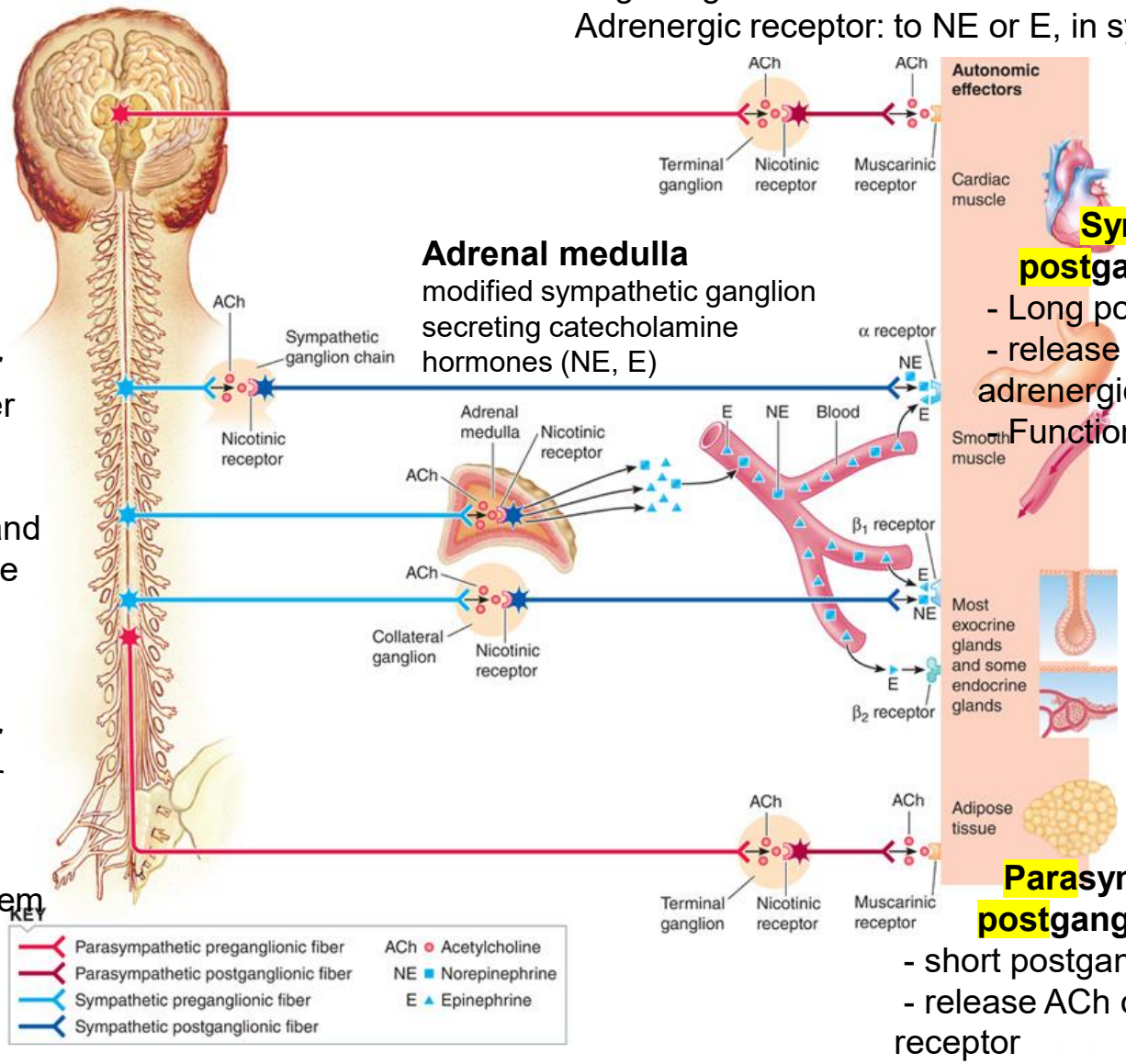
- Two autonomic nerve pathway
  - Preganglionic neuron: synapses with the cell body of postganglionic fiber in autonomic ganglion
  - Postganglionic neuron: sends axons that end on the effector organ
- Two functional subdivisions
  - Sympathetic: called adrenergic fiber, most postganglionic fibers release norepinephrine to the effector ("fight-or-flight“)
  - Parasympathetic: called cholinergic fiber, postganglionic fibers release acetylcholine to the effector (“rest-and-digest”)
  - All preganglionic fibers in both division release acetylcholine

Pathways and division of the

Nicotinic receptor: receptor proteins that respond to ACh in NMJ, ganglia, adrenal medulla
Muscarinic receptor: to ACh, in parasympathetic target organs
Adrenergic receptor: to NE or E, in sympathetic

Sympathetic preganglionic fiber
- short preganglion fiber
- release ACh onto nicotinic receptors
- originate from chest and abdominal regions of the spinal cord (T1-L2)

Parasympathetic preganglionic fiber
- long preganglion fiber
- release ACh onto nicotinic receptor
- originate from brainstem and lower spinal cord

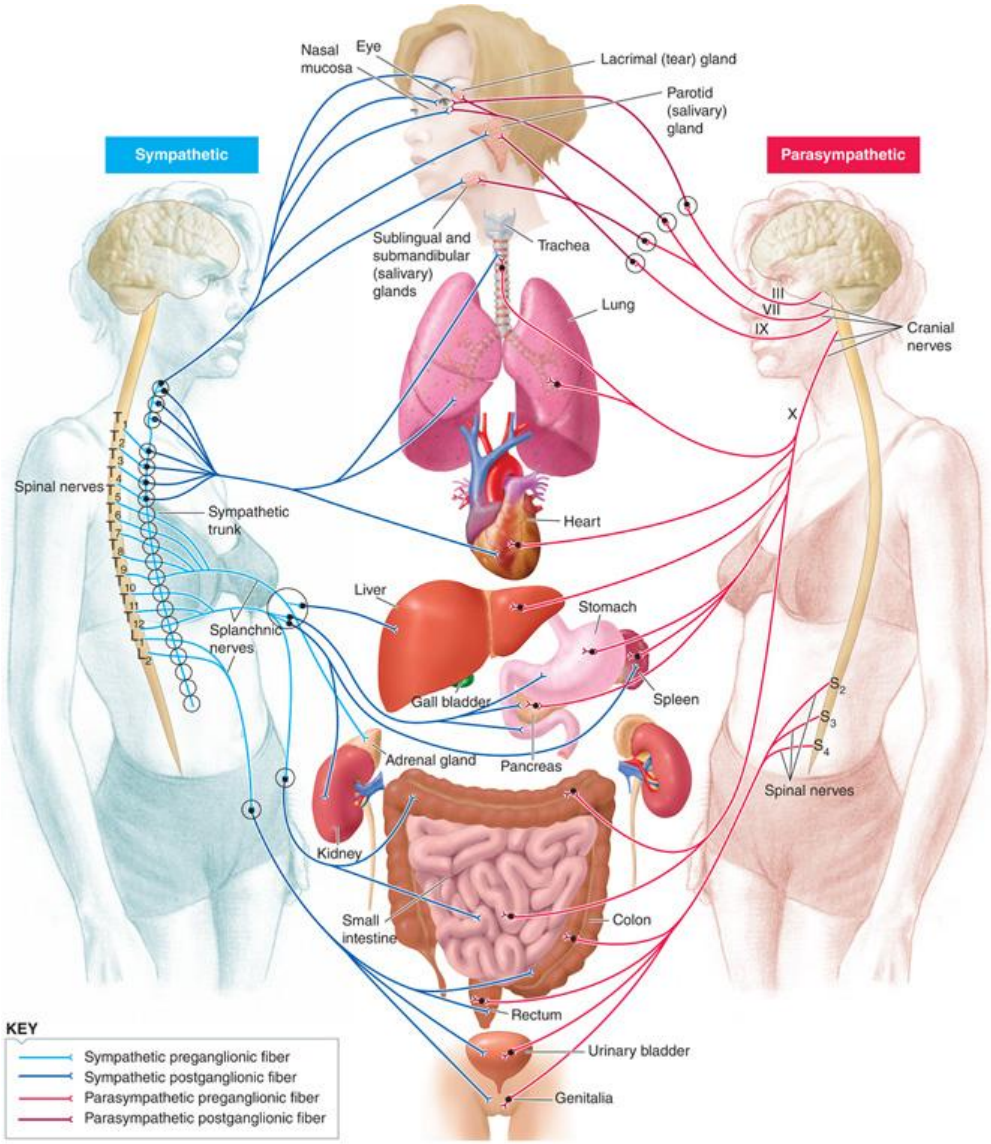


Sympathetic postganglionic fiber
- Long postganglion fiber
- release NE onto adrenergic receptor
- Function: fight or flight

Parasympathetic postganglionic fiber
- short postganglion fiber
- release ACh onto muscarinic receptor
- Function: rest & digest

Dual innervation of subdivisions

- Sympathetic and parasympathetic nervous systems dually innervate most visceral organs (내장기관)
- Dual innervation = innervation of a single organ by both branches of the ANS
- Sympathetic dominance -> “fight-or-flight”
- Parasympathetic dominance -> “rest-and-digest”
- Advantages: reciprocal control



Receptors of the ANS on effectors

- Cholinergic receptors
  - Acetylcholine-binding receptors
  - Nicotinic receptors and muscarinic receptors
- Adrenergic receptors
  - Epinephrine or norepinephrine-binding receptors
- Autonomic agonists and antagonists
  - Agonists: drugs binding to neurotransmitters' receptors and cause the same response
  - Antagonists: drug binding to the receptors and prevent NTs' binding and causing a response

Sympathetic and parasympathetic nervous system

TABLE 7-3 Comparison of the Sympathetic and the Parasympathetic Nervous System

Feature	Sympathetic System	Parasympathetic System
Origin of preganglionic fiber	Thoracic and lumbar regions of the spinal cord	Brain and sacral region of the spinal cord
Origin of postganglionic fiber	Sympathetic ganglion chain (near the spinal cord) or collateral ganglia (about halfway between spinal cord and effector organs)	Terminal ganglia (in or near effector organs)
Fiber length	Short preganglionic fibers, long postganglionic fibers	Long preganglionic fibers, short postganglionic fibers
Neurotransmitter released	Preganglionic: ACh Postganglionic: NE	Preganglionic: ACh Postganglionic: ACh
Types of receptors for neurotransmitters	For preganglionic neurotransmitter: nicotinic For postganglionic neurotransmitter: α_1, α_2, β_1, β_2	For preganglionic neurotransmitter: nicotinic For postganglionic neurotransmitter: muscarinic
Dominance	Dominates in "fight-or-flight" situations	Dominates in "rest-and-digest" situations

Many regions of the CNS controls autonomic activities

- Some autonomic reflexes (반사) are integrated and processed at the spinal-cord level
- Medullar in brain stem is directly responsible for autonomic output
- Hypothalamus integrate autonomic, somatic, and endocrine responses
- Autonomic activity can be influenced by the prefrontal association cortex (emotional expression)

7.2 Somatic Nervous System

Somatic nervous system

- Most neurons supply skeletal muscle
  - To bring about movement
  - Axons of motor neurons originate in the CNS and end on skeletal muscle
  - Motor neuron axon terminals release ACh to stimulate muscle contraction
  - Motor neurons only can stimulate, in contrast to autonomic fibers
  - Inhibition of skeletal muscle is possible only within the CNS through inhibitory synaptic input

- Motor neurons are the final common pathway
  - Only way the other nervous system can influence skeletal muscle activity is by acting on these motor neurons

Comparing ANS and PNS

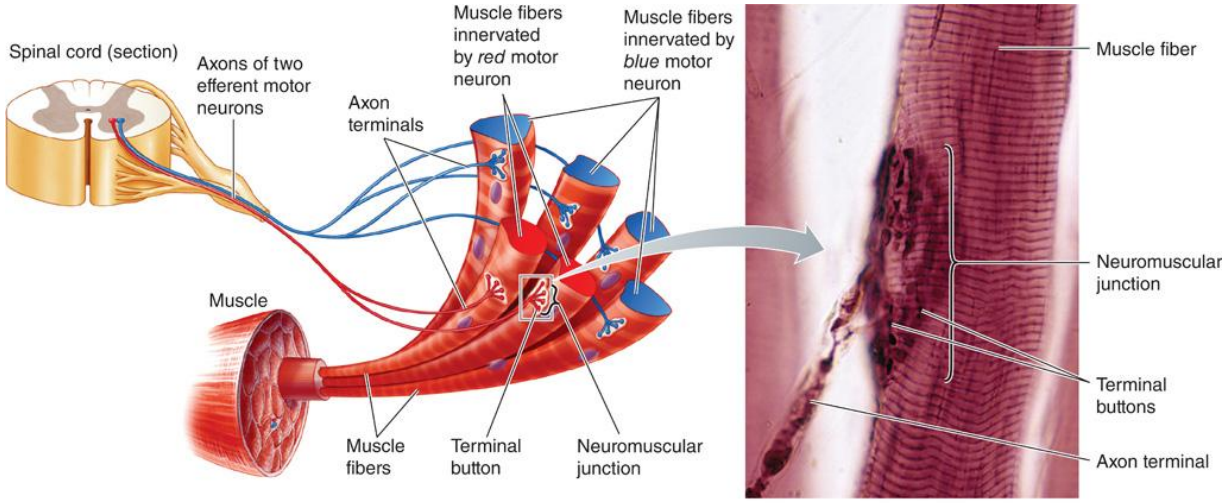
TABLE 7-4 Comparison of the Autonomic and the Somatic Nervous System

Feature	Autonomic Nervous System	Somatic Nervous System
Site of origin	Sympathetic: lateral horn of thoracic and lumbar spinal cord Parasympathetic: brain and sacral spinal cord	Ventral horn of spinal cord for most; those supplying muscles in head originate in brain
Number of neurons from CNS to effector organ	Two-neuron chain (preganglionic and postganglionic)	Single neuron (motor neuron)
Organs innervated	Cardiac muscle, smooth muscle, most exocrine and some endocrine glands	Skeletal muscle
Type of innervation	Most effector organs dually innervated by the two antagonistic branches of this system (sympathetic and parasympathetic)	Effector organs innervated only by motor neurons
Neurotransmitter at effector organs	May be ACh (parasympathetic terminals) or NE (sympathetic terminals)	Only ACh
Effects on effector organs	Either stimulation or inhibition (antagonistic actions of two branches)	Stimulation only (inhibition possible only centrally through IPSPs on dendrites and cell body of motor neuron)
Type of control	Under involuntary control	Subject to voluntary control; much activity subconsciously coordinated
Higher centers involved in control	Spinal cord, medulla, hypothalamus, prefrontal association cortex	Spinal cord, motor cortex, basal nuclei, cerebellum, brain stem

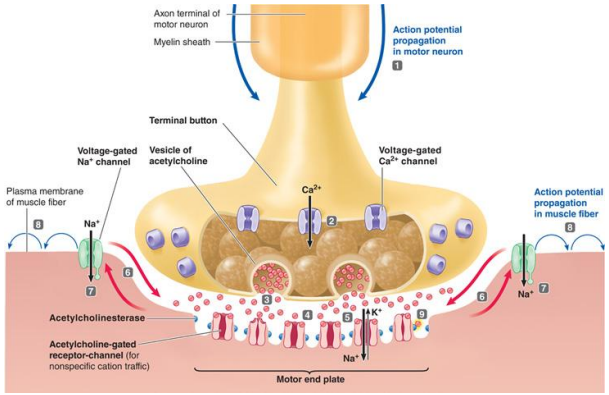
7.3 Neuromuscular Junction (NMJ)

Neuromuscular junction

- Motor neurons and skeletal muscle fibers are chemically linked at neuromuscular junction
- Each muscle cell has one neuromuscular junction
- At NMJ, the axon terminal splits into terminal button
- Motor end plate: shallow groove shaped muscle cell membrane, where axon terminals end



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Ach release at the NMJ

- Nerve and muscle cells do not come into direct contact at NMJ
  - Too large space for electrical transmission

- Like neuronal chemical synapse, neurotransmitter (acetylcholine) carries the signal between terminal button and motor end plate
- Each terminal button contains thousands of vesicles storing Ach

- 1) Propagation of AP at the axon terminal trigger 2) opening of voltage-gated Ca^{2+}-channels on terminal buttons
- 3) Ca^{2+} influx causes ACh release by exocytosis

- 4) Released ACh binds to chemically gated receptor-channels on motor end plate
- 5) Opening of these nonspecific cationic channels permit Na^{+} influx
- 6) End-plate potential (EPP), a graded potential, is generated

Ach release at the NMJ

Initiation of AP

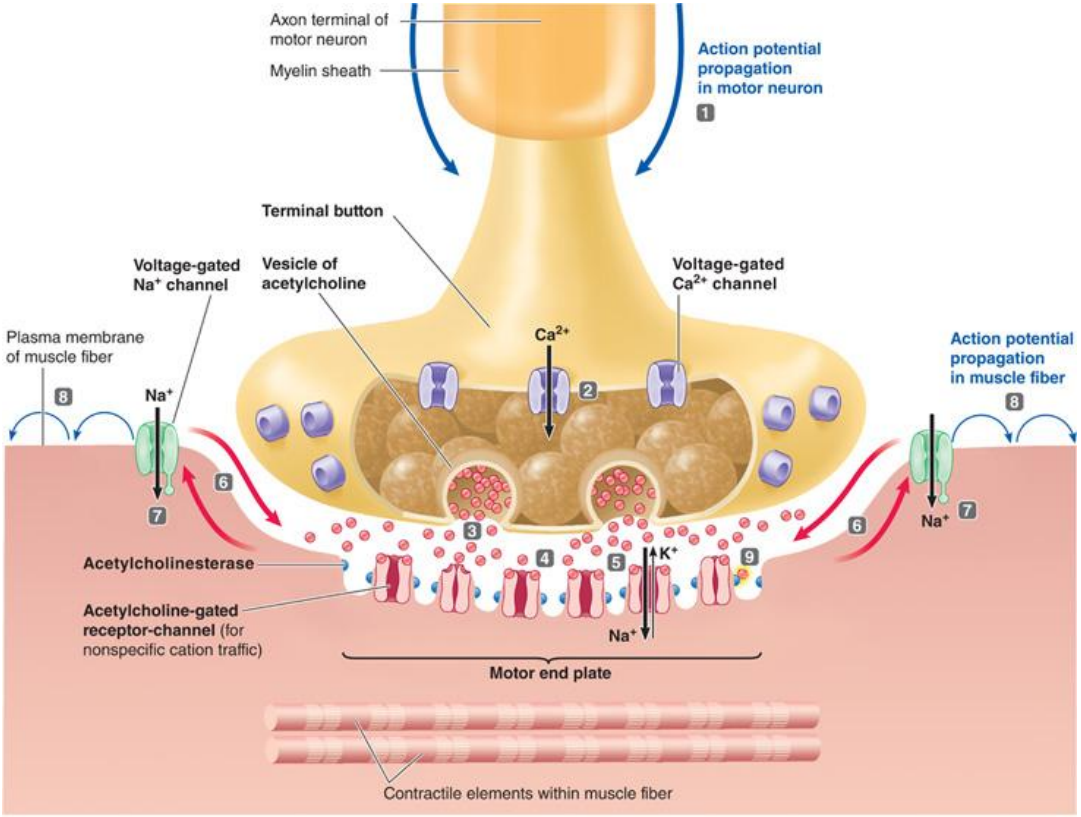
- EPP generates the *local current flow* between the depolarized end plate and the adjacent membrane

- 7) This local current flow opens voltage-gated Na^+ channels in the adjacent membrane
- 8) Additional Na^+ entry elevates membrane potential to threshold for action potential propagation throughout the muscle fiber

Termination of ACh activity

- 9) ACh is then degraded by acetylcholinerase, localized on motor end plate membrane, to terminate the response

Events at the NMJ



1 An action potential in a motor neuron is propagated to the terminal button.

2 This local action potential triggers the opening of voltage-gated Ca^{2+} channels and the subsequent entry of Ca^{2+} into the terminal button.

3 Ca^{2+} triggers the release of acetylcholine (ACh) by exocytosis from a portion of the vesicles.

4 ACh diffuses across the space separating the nerve and muscle cells and binds with receptor-channels specific for it on the motor end plate of the muscle cell membrane.

5 This binding brings about the opening of these nonspecific cation channels, leading to a relatively large movement of Na^+ into the muscle cell compared to a smaller movement of K^+ outward.

6 The result is an end-plate potential. Local current flow occurs between the depolarized end plate and the adjacent membrane.

7 This local current flow opens voltage-gated Na^+ channels in the adjacent membrane.

8 The resultant Na^+ entry reduces the potential to threshold, initiating an action potential, which is propagated throughout the muscle fiber.

9 ACh is subsequently destroyed by acetylcholinesterase, an enzyme located on the motor end-plate membrane, terminating the muscle cell's response.

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